My Project

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

Chapter 3

Class Documentation

3.1 FShop Class Reference

Public Member Functions

- FShop (std::size_t jobs, std::size_t machines, unsigned int(∗makespanfn)(std::vector< std::vector< unsigned int > > &, std::vector< std::size_t > &))
- void updateMakespan ()

Public Attributes

- std::vector< std::vector< unsigned int > > **f_shop**
- std::vector< std::size_t > **job_index_order**
- unsigned int **makespan**
- unsigned int(∗ **calcMakespan**)(std::vector< std::vector< unsigned int > > &fs, std::vector< std::size_t > &ji)

3.1.1 Constructor & Destructor Documentation

3.1.1.1 FShop()

```
FShop::FShop (
            std::size_t jobs,
            std::size_t machines,
            unsigned int(* makespanfn )(std::vector< std::vector< unsigned int > > &, std::↩
vector< std::size_t > &))
```

Constructor of FShop

Parameters

in	*jobs*	the number of jobs in a flow shop
in	*machines*	the number of machines in a flow shop

in	*makespan*	a function to calculate the makespan of a flow shop
out	*a*	flow shop initialized with every value as [job][machine] as 0

3.1.2 Member Function Documentation

3.1.2.1 updateMakespan()

```
void FShop::updateMakespan ()
```

Updates the makespan of a FShop using its makespan function

The documentation for this class was generated from the following files:

- FShop.h
- FShop.cpp

Chapter 4

File Documentation

4.1 ACO.h

```
// Name: Harrison Ingram-Bate
#ifndef ACO_H
#define ACO_H

#include "FShop.h"

FShop ACO(FShop& fs,
          std::size_t num_ants,
          int iterations,
          float alpha,
          float beta,
          float evaporation,
          float q);

#endif
```

4.2 FShop.h

```
// Name: Harrison Ingram-Bate
#ifndef F_SHOP
#define F_SHOP

#include <vector>

class FShop {
public:
    FShop(std::size_t jobs,
       std::size_t machines,
       unsigned int (*makespanfn)(std::vector<std::vector<unsigned int»&, std::vector<std::size_t>&));
    void updateMakespan();

    std::vector<std::vector<unsigned int» f_shop;
    std::vector<std::size_t> job_index_order;
    unsigned int makespan;
    unsigned int (*calcMakespan)(std::vector<std::vector<unsigned int»& fs, std::vector<std::size_t>& ji);
};

#endif
```

4.3 MakespanFunctions.h

```
// Name: Harrison Ingram-Bate
#ifndef MAKESPAN_FUNCTIONS
#define MAKESPAN_FUNCTIONS

#include <vector>

```

```
namespace makespan_functions{

    unsigned int nonBlocking(std::vector<std::vector<unsigned int»& fs, std::vector<std::size_t>& ji);

    unsigned int blocking(std::vector<std::vector<unsigned int»& fs, std::vector<std::size_t>& ji);
}

#endif
```

4.4 NEH.h

```
// Name: Harrison Ingram-Bate
#ifndef NEH_H
#define NEH_H

#include "FShop.h"

FShop NEH(FShop& fs);

#endif
```

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